

9. The Most General Domains from Integral Transcendental Numbers

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By means of the well-ordering theorem, E. Zermelo constructed a domain of “integral transcendental numbers,” that is, a numerical domain $\mathfrak{G}$ having the following abstract properties:

- I. The sum, difference, and product of two numbers from $\mathfrak{G}$ is again a number from $\mathfrak{G}$.
- II. Every real or complex number is the quotient of two numbers from $\mathfrak{G}$.
- III. Every integral rational (respectively algebraic) number is an element of $\mathfrak{G}$.
- IV. No non-integral rational (respectively algebraic) number belongs to $\mathfrak{G}$.¹

The construction rests on the fact that the well-ordering theorem implies the existence of an *algebraic basis* of all numbers: namely, a *system H of numbers η* among which no algebraic relations exist, but by means of which all remaining numbers can be expressed algebraically. Because of their algebraic independence, these basis numbers η can be regarded as indeterminates, and the desired domain therefore becomes an integral domain of rational and algebraic functions of indeterminates η ,² whose coefficients belong to the field K of all algebraic numbers. The special domain $\mathfrak{G}_\eta$ constructed by Zermelo is then characterized by the following basis properties:

$\mathfrak{G}_\eta$ contains:

- 1) all basis numbers η ,
- 2) all polynomials in the η with integral³ coefficients,
- 3) all integral algebraic functions of the η with integral coefficients.

$\mathfrak{G}_\eta$ excludes:

- 4) all rationally fractional functions of the η with integral coefficients,
- 5) all algebraically fractional functions of the η with integral coefficients.⁴

It is now easy to see (compare the examples in §§ 2 and 3) that there are domains $\mathfrak{G}$ essentially different from the Zermelo domain $\mathfrak{G}_\eta$: that is, domains $\mathfrak{G}$ for which, under no well-ordering, do all basis properties 1)–5) hold, and which consequently cannot be generated by Zermelo’s construction under any well-ordering. In other words, there are domains $\mathfrak{G}$ that cannot be mapped one-to-one and isomorphically onto $\mathfrak{G}_\eta$. Thus the question arises of the most general domains from integral transcendental numbers; it will be answered completely in what follows.

For this purpose one must first determine which of the basis properties are consequences of the abstract defining properties and which are caused by the special construction. It is shown (§ 1) that for every prescribed domain $\mathfrak{G}$ there is a well-ordering such that basis properties 1) and 2) are satisfied; hence every domain $\mathfrak{G}$ can be constructed by means of an algebraic basis of all numbers. None of the further basis properties need be satisfied (§§ 2 and 3). In particular it follows that a further abstract property of the Zermelo domain does not belong to all domains $\mathfrak{G}$: algebraic integral closedness. It can be formulated as follows:

- V. Every number that depends algebraically-integrally and integrally on finitely many numbers from $\mathfrak{G}$ belongs to $\mathfrak{G}$.

The special domains for which V holds in addition to I–IV will be called domains $\mathfrak{H}$ from algebraically-integral transcendental numbers. For the most general domains $\mathfrak{H}$ one obtains the simple results (§ 4):

¹E. Zermelo, Über ganze transzendente Zahlen, Math. Ann. 75, p. 434 (1914).

²For this reason the considerations of the present paper partly run parallel to those of the author’s earlier paper: Körper und Systeme rationaler Funktionen, Math. Ann. 76, p. 161 (1915).

³Throughout, “integral” is to mean that the coefficients belong to the integral domain $[K]$ of all algebraic integers. For the definition of $\mathfrak{G}_\eta$, it would also suffice in 1)–5) to consider only rational integer coefficients; for more general domains, however, the basis properties would then become less simple.

⁴Zermelo, loc. cit., § 3.

- a) The intersection of all domains $\mathfrak{H}$ constructible by means of the same basis H is given by all integral algebraic functions of the basis, that is, by the Zermelo domain $\mathfrak{G}_\eta$, which is itself a domain $\mathfrak{H}$. (This also gives an abstract characterization of the Zermelo domain.)
- b) Every domain $\mathfrak{H}$ arises by algebraically-integral adjunction⁵ of an arbitrary system $\mathfrak{S}(\eta)$ to $\mathfrak{G}_\eta$, where $\mathfrak{S}(\eta)$ has to satisfy only the single condition that no algebraically fractional number enters the domain through the adjunction.

The results for the most general domains $\mathfrak{G}$, when an algebraic basis is taken as foundation, are less simple. Here the intersection of all domains $\mathfrak{G}$ is itself not a domain $\mathfrak{G}$, and the construction of the most general domains is consequently harder to survey (§ 5).

In order to obtain results analogous to those for the domains $\mathfrak{H}$, the algebraic basis is replaced by a *rational basis* Θ , that is, by a *system* Θ of numbers ϑ which permits all numbers to be expressed rationally, while under at least one well-ordering no basis number can be expressed rationally by finitely many preceding ones. This rational basis Θ must still be subject to the side condition that the integral domain $[\Theta]$ consisting of all integral polynomials in the ϑ contains no algebraically fractional number. (The construction of such a basis with the side condition is shown in § 7.) Then for the most general domains $\mathfrak{G}$ — which one could also call domains from rationally integral transcendental numbers — one again obtains the simple results (§ 6):

- a) The intersection of all domains $\mathfrak{G}$ constructible by means of the same rational basis Θ is given by all integral rational functions of the basis, that is, by the domain $[\Theta]$, which is itself a domain $\mathfrak{G}$.
- b) Every domain $\mathfrak{G}$ arises by rationally integral adjunction of an arbitrary system $\mathfrak{S}(\vartheta)$ to $[\Theta]$, where $\mathfrak{S}(\vartheta)$ has to satisfy only the single condition that no algebraically fractional number enters the domain through the adjunction.

To have complete analogy with the domains $\mathfrak{H}$, one would have to omit the algebraic numbers from the defining properties III and IV for the domains $\mathfrak{G}$. With this modification of III and IV, the construction of the integral elements by means of a rational basis can be transferred to an arbitrary field (§ 10), whereas Zermelo's construction presupposes the algebraic closedness of the field.

As H runs through every algebraic basis, respectively as Θ runs through every rational basis satisfying the side condition, the results a) and b) give the totality of all domains $\mathfrak{H}$ and $\mathfrak{G}$. If all isomorphic domains are collected into a class, then from this totality one can select a subset which represents the totality of all classes — at least countably infinitely many by § 9 (§ 8).

§ 1. Proof of basis properties 1) and 2) for all domains $\mathfrak{G}$.

Let $\mathfrak{G}$ be any prescribed domain from integral transcendental numbers, so that it has the abstract properties I–IV. Following the procedure applied by E. Zermelo to the field of all complex numbers, we select from $\mathfrak{G}$ an *algebraic basis* H of all numbers from $\mathfrak{G}$. Since by II every real or complex number can be represented as the quotient of two numbers from $\mathfrak{G}$, H is at the same time an algebraic basis of *all* numbers. Thus, for every prescribed domain $\mathfrak{G}$, the algebraic basis of all numbers can be chosen so that it belongs to $\mathfrak{G}$; hence basis property 1) is satisfied. Moreover, by III, $\mathfrak{G}$ contains the integral domain $[K]$ of all algebraic integers, and therefore, by I, also all polynomials in the basis elements η with coefficients from $[K]$, that is, all integral polynomials in the η , which form the integral domain $[H]$. Hence basis property 2) is also always satisfied; in other words, every domain $\mathfrak{G}$ can be constructed by means of an algebraic basis H of all numbers in such a way that $\mathfrak{G}$ contains the integral domain $[H]$ as a divisor.

Remark. Nevertheless, starting from a basis H , one can of course construct a domain $\mathfrak{G}$ for which basis properties 1) and 2) do not hold with respect to H itself, but do hold with respect to some other basis H' . For example, let $\mathfrak{G}'$ arise from the Zermelo domain $\mathfrak{G}_\eta$ by replacing, throughout the whole construction, some basis element η by a polynomial $f(\eta)$. Then $\mathfrak{G}'$ contains neither η nor $[H]$; but if in the basis H the basis element η is replaced by $\xi = f(\eta)$ — whereby H again goes over into an algebraic basis H' —, then basis properties 1) and 2) are satisfied for $\mathfrak{G}'$ with respect to H' .

§ 2. Exclusion of basis properties 4) and 5).

We now show that the domains $\mathfrak{G}$ need not satisfy any of the further basis properties. First 4) and 5) are excluded by replacing, in Zermelo's construction, the integral domain $[H]$ by a domain of integral rational

⁵Explanation of the expression in § 4.

“functionals.”

According to Weber,⁶ a *rational functional* means every rational function with rational number coefficients in the following uniquely determined representation:

$$A(\eta_1 \cdots \eta_t) = a \cdot \frac{E_1(\eta_1 \cdots \eta_t)}{E_2(\eta_1 \cdots \eta_t)},$$

where E_1, E_2 are relatively prime primitive polynomials with rational integer coefficients, and a is a positive rational number (the absolute value of A). If, in particular, a is a *rational integer*, then A is called an *integral rational functional*. As special cases, the integral rational functionals include the rational integers and the polynomials with rational integer coefficients, while rationally fractional numbers and polynomials with rationally fractional coefficients are to be counted among the fractional rational functionals.

Let the domain $\mathfrak{G}$ now consist of the totality $\mathfrak{F}$ of all integral rational functionals of finitely many basis elements η at a time, and of all functions which depend algebraically-integrally on $\mathfrak{F}$ — that is, which satisfy some equation whose leading coefficient is unity and whose remaining coefficients are integral rational functionals.

The proof of properties I–IV for $\mathfrak{G}$ runs exactly parallel to the corresponding proof for Zermelo and will only be indicated briefly. First, II and III are clear, since $\mathfrak{G}$ contains the Zermelo domain $\mathfrak{G}_\eta$ as a divisor. By Gauss’s theorem on polynomials with rational integer coefficients, sums, differences, and products of integral rational functionals are again integral rational functionals; hence $\mathfrak{F}$ forms an integral domain, and by familiar arguments⁷ $\mathfrak{G}$ also becomes an integral domain; thus I is fulfilled. Further, from the extended Gauss theorem⁸ follow the two auxiliary propositions: “If z depends algebraically-integrally on $\mathfrak{F}$ by means of *some* equation, then it also does so by means of the *irreducible* equation which z satisfies with respect to the field of all rational functionals,”⁹ and “the equation irreducible over the field of all rational numbers and satisfied by an algebraic number is also irreducible over the field of all rational functionals.” Thus $\mathfrak{G}$ can contain no fractional algebraic number; hence IV, and therefore all abstract properties I–IV, have been verified.

On the other hand, under no well-ordering can a basis be selected from $\mathfrak{G}$ in such a way that $\mathfrak{G}$ excludes all rationally and algebraically fractional functions of the basis elements. For let $z(\eta)$ be any function of the domain to be chosen as a basis element, hence defined by an equation

$$z^k + A_1(\eta)z^{k-1} + \cdots + A_k(\eta) = 0,$$

where

$$A_i(\eta) = a_i \cdot \frac{E_1(\eta)}{E_2(\eta)}$$

are integral rational functionals. Then the function $\frac{a_k}{z}$ belongs to $\mathfrak{G}$, and likewise $\frac{a_k}{\sqrt{z}}$, $\frac{a_k}{\sqrt[3]{z}}$, etc.¹⁰ Hence basis properties 4) and 5) are excluded for the totality of the domains $\mathfrak{G}$.

Remark. An example in § 3 will show that $\mathfrak{G}$ may, under every well-ordering, contain rationally fractional functionals without containing a rationally or algebraically fractional number.

§ 3. Exclusion of basis property 3).

For excluding basis property 3), we use a generalized *module domain* in which the arguments of the polynomials of the domain are no longer the indeterminates, but all integral algebraic functions of the indeterminates,¹¹ while the indeterminates themselves take over the role of a module basis.

Let $\mathfrak{G}$ consist of *all elements*

$$z = \alpha + g_1(\eta)\eta_1 + g_2(\eta)\eta_2 + \cdots + g_t(\eta)\eta_t, \quad (1)$$

⁶H. Weber, Lehrbuch der Algebra. Kleine Ausgabe §§ 96 and 97.

⁷Compare, for example, Weber, § 97, 8.

⁸Weber, § 20, 5.

⁹Compare Weber, § 97, 4.

¹⁰Quite analogously, every algebraic function of the η can be transformed into an element of the functional domain $\mathfrak{G}$ by multiplication by a rational integer. Thus this domain has the property that every complex number can be represented as the quotient of an element from $\mathfrak{G}$ and a rational integer, in analogy with the algebraic numbers.

¹¹By “integral algebraic functions” are always meant functions with *integral* coefficients; that is, functions satisfying some equation whose leading coefficient is unity and whose remaining coefficients are polynomials in the η with algebraic-integer coefficients — polynomials from $[H]$. In particular, all polynomials from $[H]$ belong to the integral algebraic functions.

where α runs through all algebraic integers, $\eta_1 \cdots \eta_t$ through all basis elements, and for each fixed combination $\eta_1 \cdots \eta_t$, the functions $g_1(\eta) \cdots g_t(\eta)$ individually run through all integral algebraic functions of the η .¹²

It is easy to verify that properties I–IV hold for this module domain. First I holds, since the sum, difference, and product of two elements z again takes the form (1). The domain $\mathfrak{G}$ also contains all basis elements η , and besides all elements $\eta \cdot g(\eta)$, where $g(\eta)$ runs through all integral algebraic functions of the η . Hence all integral algebraic functions of the η , and therefore all algebraic functions of the η altogether, can be represented as quotients of two elements of $\mathfrak{G}$, proving II. In the representation (1), all algebraic integers are also included in particular — for $g_1 = 0 \cdots g_t = 0$ —; but z cannot be equal to a fractional algebraic number. For if z represents an algebraic number β , then from

$$\beta = \alpha + g_1(\eta)\eta_1 + \cdots + g_t(\eta)\eta_t$$

identically in the η , necessarily $\beta = \alpha$. This is shown by the specialization

$$\eta_1 = 0 \cdots \eta_t = 0,$$

under which the multipliers $g_1(\eta) \cdots g_t(\eta)$ remain finite as integral algebraic functions, since they go over into integral algebraic functions or numbers.¹³ Thus conditions I–IV are satisfied for $\mathfrak{G}$.

On the other hand, under no well-ordering can a basis be selected from $\mathfrak{G}$ in such a way that all integral algebraic functions of the basis elements belong to $\mathfrak{G}$. For let $z(\eta)$ be any function of the domain to be chosen as a basis element — which therefore must actually contain the indeterminates η —. We shall show that, from a certain finite value of ν onward, depending on z , the functions

$$\xi = \sqrt[\nu]{z - \alpha}$$

can no longer belong to $\mathfrak{G}$.

Indeed, suppose that ξ belongs to $\mathfrak{G}$. Then, by (1), it is of the form

$$\xi = \gamma + h_1(\eta)\eta_{i_1} + h_2(\eta)\eta_{i_2} + \cdots + h_\tau(\eta)\eta_{i_\tau},$$

where again $h_1(\eta) \cdots h_\tau(\eta)$ are integral algebraic functions of the η , and $\eta_{i_1} \cdots \eta_{i_\tau}$ are arbitrary basis elements, which in particular may also coincide with $\eta_1 \cdots \eta_t$. First it must be shown that the algebraic integer γ is zero. Since $h_1(\eta) \cdots h_\tau(\eta)$ are integral algebraic functions, ξ becomes equal to γ for $\eta_{i_1} = 0 \cdots \eta_{i_\tau} = 0$ with arbitrary values of the remaining η ; in particular, then, ξ is equal to γ for

$$\eta_{i_1} = 0 \cdots \eta_{i_\tau} = 0; \quad \eta_1 = 0 \cdots \eta_t = 0.$$

But under this specialization the function $(z - \alpha)$ vanishes by (1), and therefore so does ξ ; hence $\gamma = 0$, and one has

$$\xi = h_1(\eta)\eta_{i_1} + h_2(\eta)\eta_{i_2} + \cdots + h_\tau(\eta)\eta_{i_\tau}.$$

Raising this to the ν -th power gives

$$\xi^\nu = z - \alpha = F_\nu(\eta),$$

where $F_\nu(\eta)$ denotes a homogeneous, not identically vanishing, form of ν -th dimension in $\eta_{i_1} \cdots \eta_{i_\tau}$, whose coefficients are integral algebraic functions of the η . Now $z - \alpha$, which by (1) is an integral algebraic function of the η , satisfies an equation irreducible with respect to $[H]$:

$$(z - \alpha)^x + B(\eta)(z - \alpha)^{x-1} + \cdots + C(\eta) = 0,$$

where the last coefficient $C(\eta)$ — the product of $(z - \alpha)$ with its conjugates — is a polynomial; let its degree be λ . On the other hand, from $z - \alpha = F_\nu(\eta)$, by multiplying $F_\nu(\eta)$ by the corresponding conjugates, one obtains

$$C(\eta) = G_{\nu\chi}(\eta),$$

where $G_{\nu\chi}(\eta)$ denotes a homogeneous form of $\nu\chi$ -th dimension in $\eta_{i_1} \cdots \eta_{i_\tau}$, whose coefficients are polynomials in the η . Thus $G_{\nu\chi}$ is at least of degree $\nu\chi$ in η , or $\lambda \geq \nu\chi$; that is, the functions $\sqrt[\nu]{z - \alpha}$ for which $\nu > \lambda/\chi$ do not belong to $\mathfrak{G}$. Basis property 3) is therefore excluded for the totality of the domains $\mathfrak{G}$.

¹²The module property belongs to the elements $z - \alpha$.

¹³This more detailed proof of IV is given because of the expanded example at the end of the paragraph. Here the remark would suffice that, by construction, z is an integral algebraic function.

Remark. A slight generalization of the domain just constructed leads to a domain $\mathfrak{G}$ which, under every well-ordering, also contains fractional functionals. Let $\mathfrak{G}$ consist of all elements

$$z = \alpha + \gamma_1(\eta)\eta_1 + \gamma_2(\eta)\eta_2 + \cdots + \gamma_t(\eta)\eta_t,$$

where now $\gamma(\eta)$ runs through all algebraically integral, but not necessarily integral, functions of the η . The proof of properties I–IV remains the same as above. But since $\mathfrak{G}$, besides every function z , also contains $a \cdot (z - \alpha)$, where a denotes any rational or algebraically fractional number, fractional functionals also occur under every well-ordering.

§ 4. The most general domains from algebraically-integral transcendental numbers.

For a more convenient formulation of what follows, introduce the expression: “an element z depends algebraically-integrally on $\mathfrak{T}$,” meaning that z satisfies some equation whose leading coefficient is unity and whose remaining coefficients are integral polynomials in the elements of $\mathfrak{T}$, where $\mathfrak{T}$ denotes any prescribed domain.¹⁴

A domain $\mathfrak{T}$ is called *algebraically-integrally closed* if it satisfies the condition (compare the introduction):

V. Every element that depends algebraically-integrally on $\mathfrak{T}$ belongs to $\mathfrak{T}$.

We first show that the abstract property V belongs to the Zermelo domain $\mathfrak{G}_\eta$. Let an element z depend algebraically-integrally on $\mathfrak{G}_\eta$ by means of an equation $f(z) = 0$. Then multiplication of $f(z)$ by its conjugates with respect to $[H]$ shows that z also depends algebraically-integrally on $[H]$, and hence belongs to $\mathfrak{G}_\eta$. The algebraically-integral closedness of the functional domain considered in § 2 is shown in the same way.

That property V does not belong to every domain $\mathfrak{G}$ is shown by the module domain considered in § 3, which indeed does not possess basis property 3). More precisely, § 3 showed that the module domain is algebraically-integrally closed with respect to no transcendental element of the domain, whereas, by III and IV, this is of course the case with respect to every algebraic element of the domain.

This leads us first to select, from the totality of domains $\mathfrak{G}$, those which also possess the abstract property V; these are to be called “domains $\mathfrak{H}$ from algebraically-integral transcendental numbers.”

Let $\mathfrak{H}$ be such a domain, and let H be an algebraic basis of all numbers from $\mathfrak{H}$; by II, H is at the same time an algebraic basis of all numbers. By III, I, and V, $\mathfrak{H}$ contains, besides H , all integral algebraic functions of the η , that is, the Zermelo domain $\mathfrak{G}_\eta$. Thus, in analogy with § 1: every domain $\mathfrak{H}$ from algebraically-integral transcendental numbers can be constructed by means of an algebraic basis of all numbers in such a way that $\mathfrak{H}$ contains the Zermelo domain $\mathfrak{G}_\eta$ as a divisor.

Now let H be any algebraic basis of all numbers, and let $\mathfrak{M}(H)$ denote the totality of all domains $\mathfrak{H}$ containing H . Every domain $\mathfrak{H}$ in $\mathfrak{M}(H)$ contains, as just proved, the Zermelo domain $\mathfrak{G}_\eta$ as a divisor; and since $\mathfrak{G}_\eta$ itself is a domain $\mathfrak{H}$, $\mathfrak{G}_\eta$ is the greatest common divisor, or intersection, of all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$. Thus $\mathfrak{G}_\eta$ has been defined abstractly. It also follows directly that $\mathfrak{M}(H)$ is closed under intersection: that is, the intersection of all domains $\mathfrak{H}$ of any subsystem of $\mathfrak{M}(H)$ again belongs to $\mathfrak{M}(H)$. For, as an intersection, I and V are satisfied; II and III follow because $\mathfrak{G}_\eta$ is a divisor; IV follows because the intersection is a divisor of $\mathfrak{H}$.

As the example of the functional domain in § 2 shows, the domains $\mathfrak{H}$ in general contain further functions besides $\mathfrak{G}_\eta$. Let $\psi(\eta)$ be such a rational or algebraic function of the η . Then, by I, $\mathfrak{H}$ also contains all integral rational combinations of ψ and finitely many functions from $\mathfrak{G}_\eta$ at a time. The integral domain so arising — consisting of all polynomials in ψ with coefficients from $\mathfrak{G}_\eta$ — will be denoted by $[\mathfrak{G}_\eta, \psi]$, and the process leading to it will be called “rationally integral adjunction of ψ to $\mathfrak{G}_\eta$.” By V, $\mathfrak{H}$ further contains all functions which depend algebraically-integrally on $[\mathfrak{G}_\eta, \psi]$; the domain so arising will be denoted by $\{\mathfrak{G}_\eta, \psi\}$, and the process leading to it by “algebraically integral adjunction of ψ to $\mathfrak{G}_\eta$.” The domain $\{\mathfrak{G}_\eta, \psi\}$ consists of all integral algebraic functions of ψ with coefficients from $\mathfrak{G}_\eta$, and is therefore an integral domain by familiar arguments.¹⁵

We show that $\{\mathfrak{G}_\eta, \psi\}$ is also algebraically-integrally closed, that is, satisfies condition V. Indeed, let z depend algebraically-integrally on $\{\mathfrak{G}_\eta, \psi\}$ by means of an equation

$$f(z; a \cdots c) = z^\nu + az^{\nu-1} + \cdots + c = 0,$$

¹⁴For an integral domain $\mathfrak{T}$, the leading coefficient is unity and the remaining coefficients are elements of $\mathfrak{T}$.

¹⁵Compare, for example, Weber, § 97, 6 and 7.

where $a \cdots c$, as elements of $\{\mathfrak{G}_\eta, \psi\}$, satisfy equations with coefficients from $[\mathfrak{G}_\eta, \psi]$:

$$\begin{aligned} a^\lambda + A_1 a^{\lambda-1} + \cdots + A_\lambda &= 0, \\ &\vdots \\ c^\mu + C_1 c^{\mu-1} + \cdots + C_\mu &= 0. \end{aligned}$$

Let their roots be denoted by $a, a' \cdots a^{(\lambda-1)} \cdots c, c' \cdots c^{(\mu-1)}$. If one now forms the product over all combinations of the roots,

$$g(z) = f(z; a \cdots c) f(z; a' \cdots c') \cdots f(z; a^{(\lambda-1)} \cdots c^{(\mu-1)}),$$

then $g(z)$ has unity as leading coefficient and elements of $[\mathfrak{G}_\eta, \psi]$ as its remaining coefficients. Consequently z belongs to $\{\mathfrak{G}_\eta, \psi\}$.¹⁶

Thus the domain $\{\mathfrak{G}_\eta, \psi\}$ has properties I and V; and, since $\mathfrak{G}_\eta$ is a divisor of the domain, it also has II and III. Whether IV is satisfied depends on the choice of ψ ; for example, IV is not satisfied for $\psi = \frac{1}{n\eta}$, since then $\frac{1}{n}$ belongs to the domain. It is satisfied, however, by § 2 for $\psi = \frac{1}{\eta}$, or also for $\psi = \frac{\eta}{n}$. For in the latter case, if a function from $\{\mathfrak{G}_\eta, \psi\}$ represents an algebraic number, its value remains unchanged for $\eta = 0$; thereby it passes into a function from $\mathfrak{G}_\eta$, and consequently into an algebraic integer.

Thus $\{\mathfrak{G}_\eta, \psi\}$ becomes a domain $\mathfrak{H}$ as soon as one imposes on ψ the condition that no algebraically fractional number enters the domain through the algebraically integral adjunction.

Quite analogously one defines the rationally integral, respectively algebraically integral, adjunction of an arbitrary system $\mathfrak{S}$ of rational or algebraic functions of the η to $\mathfrak{G}_\eta$. The rationally integral adjunction leads to the integral domain $[\mathfrak{G}_\eta, \mathfrak{S}]$, consisting of all integral polynomials in finitely many elements from $\mathfrak{G}_\eta$ and $\mathfrak{S}$ at a time. The algebraically integral adjunction gives the domain $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ consisting of all elements that depend algebraically-integrally on $[\mathfrak{G}_\eta, \mathfrak{S}]$. Exactly as above, one shows that the integral domain $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ is algebraically-integrally closed and also satisfies II and III. Thus, as above:

$\{\mathfrak{G}_\eta, \mathfrak{S}\}$ becomes a domain $\mathfrak{H}$ as soon as one imposes on $\mathfrak{S}$ the condition that no algebraically fractional number enters the domain through the algebraically integral adjunction — that is, as soon as $\mathfrak{S}$ becomes an admissible system.¹⁷

It is now important that the converse also holds, so that the domains $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ actually exhaust all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$ as $\mathfrak{S}$ runs through all admissible systems. Indeed, let $\mathfrak{H}$ be any domain from $\mathfrak{M}(H)$ and denote by $\mathfrak{L} = \mathfrak{H} - \mathfrak{G}_\eta$ the system consisting of all functions from $\mathfrak{H}$ not belonging to $\mathfrak{G}_\eta$. By V, $\mathfrak{H}$ contains the domain $\{\mathfrak{G}_\eta, \mathfrak{L}\}$ as a divisor; but $\{\mathfrak{G}_\eta, \mathfrak{L}\}$ is also divisible by $\mathfrak{H}$, because it contains $\mathfrak{G}_\eta$ and $\mathfrak{L}$, and since these have no common element, it contains $\mathfrak{H}$ as well. Hence $\mathfrak{H} = \{\mathfrak{G}_\eta, \mathfrak{L}\}$; and since IV is satisfied for $\mathfrak{H}$, $\mathfrak{L}$ is of course an admissible system.¹⁸

As H runs through every possible algebraic basis, $\mathfrak{M}(H)$ runs, as proved at the beginning, through the totality of all domains $\mathfrak{H}$. Consequently the question of the most general domains $\mathfrak{H}$ from algebraically-integral transcendental numbers is completely answered by the results of this paragraph:

- a) The intersection of all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$ is given by the Zermelo domain $\mathfrak{G}_\eta$, which itself is a domain $\mathfrak{H}$; $\mathfrak{M}(H)$ is closed with respect to forming intersections.
- b) Every domain $\mathfrak{H}$ from $\mathfrak{M}(H)$ arises by algebraically integral adjunction of an admissible system $\mathfrak{S}$ to $\mathfrak{G}_\eta$. As H runs through every possible algebraic basis, $\mathfrak{M}(H)$ runs through the totality of all domains $\mathfrak{H}$.¹⁹

¹⁶Thus, by defining algebraic integrality by means of *some* equation, the concept of irreducibility with respect to a ground field becomes unnecessary for the proof of I and V. Compare also § 2, where, only for the proof of IV, the auxiliary proposition is needed which passes from the definition of algebraic integrality by an arbitrary equation to the irreducible equation. Since this auxiliary proposition no longer holds in general in the present case, IV has to be imposed on the function ψ as a special condition.

¹⁷More generally, one shows analogously: if $\mathfrak{T}$ is any domain and $\{\mathfrak{T}\}$ denotes the totality of elements depending algebraically-integrally on $\mathfrak{T}$, then $\{\mathfrak{T}\}$ is algebraically-integrally closed.

¹⁸The adjunction of a subsystem of $\mathfrak{L}$ can already suffice; for example, the functional domain of § 2 arises by algebraically integral adjunction of all integral rational functionals — with the exception of the polynomials — to $\mathfrak{G}_\eta$.

¹⁹The admissible systems $\mathfrak{S}$ can be found as follows. Put all algebraically fractional functions of the η under some well-ordering Ω , and take into $\mathfrak{S}$ all functions except those which, when algebraically-integrally adjoined to $\mathfrak{G}_\eta$ and the previously admitted functions, generate an algebraically fractional number. This procedure can be broken off at any arbitrary place. As Ω runs through every well-ordering, and as the process is broken off at every arbitrary place for each fixed Ω , all admissible systems $\mathfrak{S}$ are thereby exhausted.

§ 5. The most general domains made from integral transcendental numbers, on the basis of an algebraic basis.

Let H be any algebraic basis of all numbers, and let $\mathfrak{M}(\mathfrak{G})$ denote the totality of all domains $\mathfrak{G}$ which contain H and therefore $[H]$. As H runs through every possible basis, $\mathfrak{M}(\mathfrak{G})$ runs through the totality of all domains $\mathfrak{G}$; hence, as in § 4, we may restrict ourselves to considering the domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$.

The domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ all contain $[H]$ as a divisor; but, since $[H]$ itself is not a domain $\mathfrak{G}$, one cannot conclude as in § 4 that $[H]$ is the greatest common divisor. That this is nevertheless the case is shown by a countable set of domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$, obtained by a slight generalization from the module domain of § 3, and in fact having no element in common besides $[H]$.

For every fixed ν , let the domains $\mathfrak{G}_\nu$ ($\nu = 1, 2, \dots$ in inf.) consist of all quantities

$$z = f(\eta) + g_1(\eta)\eta_1^\nu + g_2(\eta)\eta_2^\nu + \dots + g_t(\eta)\eta_t^\nu, \quad (1)$$

where $f(\eta)$ runs through all integral polynomials in the η ; $\eta_1, \dots, \eta_t$ run through the basis quantities, and, for each fixed combination $\eta_1^\nu, \dots, \eta_t^\nu$, the functions $g_1(\eta), \dots, g_t(\eta)$ run independently through all algebraic integers functions of the η .

As in § 3, one sees that the domains $\mathfrak{G}_\nu$ have the abstract properties I–IV; $\mathfrak{G}_\nu$ contains only algebraic-integer functions and, for $\nu = 1$, is identical with the domain of § 3. We now show that the domains $\mathfrak{G}_\nu$ have no further common element besides the polynomials $f(\eta)$ from $[H]$. Indeed, let z be any algebraic-integer — but not rational — function of the η , satisfying the equation, irreducible with respect to $[H]$,

$$z^\chi + a_1(\eta)z^{\chi-1} + a_2(\eta)z^{\chi-2} + \dots + a_\chi(\eta) = 0, \quad (2)$$

where $\chi > 1$; and let λ be the greatest of the degrees of the polynomials $a_1(\eta), \dots, a_\chi(\eta)$. If z belongs to $\mathfrak{G}_\nu$, then the quantity $\xi = z - f(\eta)$ satisfies the likewise irreducible equation, again with respect to $[H]$,

$$\xi^\chi + b_1(\eta)\xi^{\chi-1} + b_2(\eta)\xi^{\chi-2} + \dots + b_\chi(\eta) = 0, \quad (3)$$

where, by (1), $b_1(\eta), b_2(\eta), \dots, b_\chi(\eta)$, as symmetric functions of the ξ , contain as terms of lowest dimension terms of at least the dimensions $\nu, 2\nu, \dots, \chi\nu$. Comparing (2) and (3) gives

$$b_1(\eta) = a_1(\eta) + \chi f(\eta).$$

If, therefore, the quantity $\xi = z + \frac{a_1(\eta)}{\chi}$ satisfies

$$\xi^\chi + c_2(\eta)\xi^{\chi-2} + \dots + c_\chi(\eta) = 0, \quad (4)$$

then

$$\xi - \zeta = \frac{b_1(\eta)}{\chi}; \quad c_i(\eta) \equiv b_i(\eta) \pmod{\frac{b_1(\eta)}{\chi}}, \quad (5)$$

where $b_1(\eta)$ begins with terms of at least dimension ν . Now the $c_i(\eta)$, as the comparison of (2) with (4) shows, are of degree χ in the coefficients $a_i(\eta)$ of (2), hence of degree at most $\chi\lambda$ in the η ; on the other hand, all the $b_i(\eta)$ begin with terms of at least dimension ν . Thus, if one chooses $\nu > \chi\lambda$, (5) gives

$$c_i(\eta) = 0; \quad \xi^\chi = 0 \quad \text{or} \quad \left(z + \frac{a_1(\eta)}{\chi}\right)^\chi = 0.$$

Thus z , contrary to the assumption, becomes rational in the η . Hence:

If z is an algebraic-integer, non-rational function of the η , then z can belong to no domain $\mathfrak{G}_\nu$ for which $\nu > \chi\lambda$, or:

The intersection of all domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ is given by the integral domain $[H]$, which itself is not a domain $\mathfrak{G}$; consequently $\mathfrak{M}(\mathfrak{G})$ is not closed with respect to forming intersections.

Consequently the construction of the most general domains $\mathfrak{G}$ by means of an algebraic basis is also harder to survey than that of the domains $\mathfrak{H}$. The domain $[H]$ has properties I, III and IV; if one adjoins any system $\mathfrak{G}$ rationally-integrally, then for the resulting integral domain $[H, \mathfrak{G}]$ the properties I and III remain valid, whereas IV forms a special condition on $\mathfrak{G}$. In order that a domain $\mathfrak{G}$ arise, II must also be imposed on $\mathfrak{G}$ as a condition; or, expressed differently: for every field $\mathfrak{K}$, finite over $(H)^{20}$, there must be a field $\mathfrak{L}$ over

²⁰ (H) means the field consisting of all rational functions of the η with algebraic-number coefficients.

$\mathfrak{K}$, finite over $\mathfrak{K}$, such that $[H, \mathfrak{S}]$ contains a primitive element from $\mathfrak{L}$.²¹ Conversely, every domain $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ can also be represented in the form $[H, \mathfrak{S}]$; hence one obtains all domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ when $\mathfrak{S}$ runs through all systems restricted in this way. The structure of the systems $\mathfrak{S}$ will appear more simply in the next paragraph by means of the “rational basis”; at the same time this gives a complete analogy with the theorems of § 4.

§ 6. The most general domains made from integral transcendental numbers, on the basis of a rational basis.

In analogy with the algebraic basis, the rational basis is to be defined as follows: “A system Θ of numbers ϑ is called a rational basis of all numbers if Θ permits every number to be expressed rationally — with coefficients in K ²² — while, for at least one well-ordering, no basis number can be expressed rationally — with coefficients in K — by finitely many preceding basis numbers.”²³

Let a domain $\mathfrak{G}$ made from integral transcendental numbers be subjected to a well-ordering Ω . Then it can happen that a quantity z from $\mathfrak{G}$ is rationally expressible by finitely many preceding quantities from $\mathfrak{G}$; that is, it satisfies an equation $z = \varphi(\xi)$, where $\varphi(\xi)$ is a rational function, with coefficients in K , of $\xi_1, \dots, \xi_t$. In particular, every algebraic number α of the domain satisfies the equation $z = \alpha$. The remaining quantities ϑ from $\mathfrak{G}$ form a system Θ which is to be shown to be a rational basis of all quantities from $\mathfrak{G}$. For by assumption each quantity ϑ is rationally independent of the preceding quantities from Θ . The induction argument²⁴ further shows that every relation $z = \varphi(\xi)$ entails a relation $z = \psi(\vartheta)$; for otherwise there would have to be a first element $z_0 = \varphi_0(\xi_1, \dots, \xi_t)$ which cannot be expressed rationally by the ϑ , while the preceding $\xi_1, \dots, \xi_t$ are rational functions of the ϑ ; this gives a contradiction. By II, Θ is at the same time a rational basis of all numbers; by III and I, besides Θ , $\mathfrak{G}$ contains as a divisor the integral domain $[\Theta]$ consisting of all integral polynomials in the ϑ , whereas by IV $[\Theta]$ contains no algebraically fractional number. Thus the rational basis Θ of all numbers satisfies the side condition that the integral domain $[\Theta]$ derived from Θ contains no algebraically fractional number;²⁵ Θ is a *rational basis with side condition* for all numbers. Thus, in analogy with § 2: *Every domain $\mathfrak{G}$ can be constructed by means of a rational basis with side condition, in such a way that $\mathfrak{G}$ contains the integral domain $[\Theta]$ as a divisor.*

Let Θ be any rational basis with side condition for all numbers — whose existence is therefore guaranteed by the existence of the domains $\mathfrak{G}$. Let $\mathfrak{N}(\mathfrak{G})$ denote the totality of all domains $\mathfrak{G}$ which contain Θ and therefore $[\Theta]$. As Θ runs through every possible basis with side condition, $\mathfrak{N}(\mathfrak{G})$ runs, by the above, through the totality of all domains $\mathfrak{G}$; hence we may again restrict ourselves to the domains $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$.

Now $[\Theta]$ itself is a domain $\mathfrak{G}$; for every number can be expressed rationally by Θ , and therefore as a quotient of two polynomials from $[\Theta]$; and by its construction $[\Theta]$ also satisfies the remaining conditions I, III, IV. Thus $[\Theta]$ is the greatest common divisor, or intersection, of all domains $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$; moreover $\mathfrak{N}(\mathfrak{G})$ is closed with respect to intersection; that is, the intersection of all domains $\mathfrak{G}$ of any subsystem of $\mathfrak{N}(\mathfrak{G})$ also belongs to $\mathfrak{N}(\mathfrak{G})$. For I holds for the intersection; II and III follow from the fact that $[\Theta]$ is a divisor; and IV from the fact that the intersection domain is contained as a divisor in a domain $\mathfrak{G}$.

The rationally integral adjunction of any system $\mathfrak{S}$ of rational functions of the ϑ to $[\Theta]$ — every algebraic function of the ϑ can, by the property of the rational basis, be expressed rationally through certain other ϑ — gives an integral domain $[\Theta, \mathfrak{S}]$ having properties I, II, III, and hence becomes a domain $\mathfrak{G}$ as soon as $\mathfrak{S}$ is an “admissible system,” i.e. as soon as it satisfies the condition that no algebraically fractional number enters the domain through the rationally integral adjunction. Conversely, every domain $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$ permits the representation $[\Theta, \mathfrak{S}]$, provided only that $\mathfrak{S}$ denotes the residual system $\mathfrak{G} - [\Theta]$. Thus, for the most general domains $\mathfrak{G}$, in complete analogy with § 4, one has:

a) *The intersection of all domains $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$ is given by the integral domain $[\Theta]$, which itself is a*

²¹Cf. § 7.

²²By rational functions one should always understand such functions with algebraic-number coefficients; these algebraic-number coefficients are included in the definition because of III and IV. It would be more appropriate to require rational-number coefficients both in III and IV and in the definition of the rational basis. The arguments of §§ 6 and 7 remain literally unchanged if one merely replaces the field K of all algebraic numbers by the field of all rational numbers.

²³In contrast to the linear and algebraic bases, the ordering of the elements plays a role here. For example, in the ordering $\eta, \sqrt[3]{\eta}$ both elements have to be admitted, whereas in the reverse ordering only $\sqrt[3]{\eta}$ has to be admitted.

²⁴Cf. Zermelo, loc. cit., § 1.

²⁵That this is indeed a condition is shown, for instance, by the fact that the two quantities $\eta, \frac{1}{\sqrt[3]{\eta}}$ are not admissible as basis elements, whereas $\eta, \frac{1}{\sqrt{\eta}}$ are.

domain $\mathfrak{G}$; $\mathfrak{N}(\mathfrak{G})$ is closed with respect to forming intersections.

- b) Every domain $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$ arises by rationally integral adjunction of an admissible system $\mathfrak{S}$ to $[\Theta]$. As Θ runs through every possible rational basis with side condition, $\mathfrak{N}(\mathfrak{G})$ runs through the totality of all domains $\mathfrak{G}$.²⁶

Remark. If one extracts from Θ an algebraic basis H of all quantities from Θ , then H is at the same time an algebraic basis of all numbers; conversely, every algebraic basis can be extended to a rational one by putting H first in the well-ordering. Accordingly, the construction in § 5 of the domains $\mathfrak{G}$ by means of an algebraic basis can be interpreted as follows. Decompose the system $\mathfrak{S}$ to be adjoined to $[H]$ into two subsystems $\mathfrak{S}_1$ and $\mathfrak{S}_2$; the adjunction of $\mathfrak{S}_1$ amounts to extending H to a rational basis with side condition, to which an admissible system $\mathfrak{S}_2$ is then still rationally-integrally adjoined.

§ 7. Construction of the most general rational basis with side condition.

The results of the preceding paragraph still need a supplement, since there the *existence* of a rational basis with side condition for all numbers was proved, but not a method for constructing it from a prescribed well-ordering. On account of the side condition, this construction takes a more complicated form when, instead of a domain $\mathfrak{G}$, the field of all complex numbers is present.²⁷

Let the field of all complex numbers be subjected to a well-ordering Ω . Then quantities of three kinds can occur:

1. Quantities y whose rationally integral adjunction to the preceding basis quantities — defined recursively in 3. — forces an algebraically fractional number to enter the resulting integral domain; this domain consists of all integral polynomials in y if no basis quantity precedes it. In particular, all algebraically fractional numbers are quantities y , since the resulting integral domain contains the quantities $y = \beta$.
2. Quantities z which do not have property 1, but which can be expressed rationally by finitely many preceding numbers different from the y ; thus they satisfy a relation $z = \varphi(\xi_1, \dots, \xi_t)$, where φ is a rational function with coefficients in K and none of the ξ are y . In particular, all algebraic integer numbers α belong here, since property 1 does not hold for them and they satisfy the relation $z = \alpha$.
3. Quantities ϑ for which neither 1 nor 2 holds and which are to be called basis quantities. In particular, the first transcendental number ϑ_0 in the well-ordering is such a basis quantity; for the integral polynomials in ϑ_0 cannot represent an algebraically fractional number, and ϑ_0 also cannot be rationally dependent on the preceding algebraic numbers.

The system Θ of all basis quantities ϑ is now to be shown to be a rational basis with side condition for all numbers.

Since no quantity y occurs among the ϑ , the integral domain $[\Theta]$ cannot contain any algebraically fractional number; hence the side condition is satisfied. Since, furthermore, no quantity z occurs among the ϑ , each ϑ is rationally independent of the preceding basis quantities. The induction argument shows, exactly as in § 6, that every relation $z = \varphi(\xi)$ — where no y occurs among the ξ — entails a relation $z = \psi(\vartheta)$, so that all quantities z are rationally expressible by the ϑ .

It remains only to prove the more difficult assertion that the y are also rationally expressible by the ϑ . First we show that the y depend algebraically on the ϑ . By assumption, for each y there exists at least one algebraically fractional number β admitting the representation

$$\beta = f_0(\vartheta) + f_1(\vartheta)y + \dots + f_\chi(\vartheta)y^\chi, \quad (1)$$

where $f_0(\vartheta), \dots, f_\chi(\vartheta)$ are polynomials from $[\Theta]$ and hence cannot represent algebraically fractional numbers. If y were algebraically independent of the ϑ , then (1) would hold identically in y and one would have $\beta = f_0(\vartheta)$, contradicting the properties of $[\Theta]$.

²⁶For the most general admissible systems $\mathfrak{S}$, what was said in § 4 applies, with “rational” replacing “algebraic.”

²⁷In § 6 b) it would suffice to let Θ run only through those special rational bases which consist of an algebraic basis and algebraically integral functions of the basis quantities; then the side condition is automatically satisfied. To see this, one need only, by § 6, extend an algebraic basis from $\mathfrak{G}$ to a rational one and, in the resulting basis, multiply each algebraically fractional function of the η by a suitably chosen polynomial in the η , whereby the basis property is preserved. This construction transfers unchanged to the field of all complex numbers. The question arising from § 6 concerning the most general rational basis with side condition, however, also has independent interest.

In order to infer rational dependence from algebraic dependence, we extract from Θ an algebraic basis H of all quantities from Θ . Then every y , as an algebraic function of the ϑ , is also algebraically dependent on the η ; through $y = y(\eta)$ an algebraic field $(H, y(\eta))$ over (H) is therefore determined. Since all η are at the same time quantities ϑ , the proof of rational dependence amounts to showing that for every such field there are primitive elements which are rational functions of the ϑ ; more precisely, it will be shown that $y(\eta)$ loses property 1 after multiplication by a polynomial in the η , and consequently passes into such an element.

It is known that between (H) and (H, y) there are only finitely many intermediate fields. Let $\Omega_0 = (H), \Omega_1, \dots, \Omega_\sigma$ be those among them which possess a primitive element rationally expressible by the ϑ , so that every element of the field becomes a rational function of the ϑ ; this condition is at least always satisfied for $\Omega_0 = (H)$. Let the primitive function belonging to Ω_i be given by $\frac{g_i(\vartheta)}{h_i(\vartheta)}$, where g_i, h_i are polynomials from $[\Theta]$. Then — with α_i denoting a suitably chosen integer — the polynomial from $[\Theta]$

$$\xi_i = g_i(\vartheta) + \alpha_i h_i(\vartheta)$$

is a primitive function of Ω_i , or of an algebraic field over Ω_i ,²⁸ and every function $\psi(\eta)$ from Ω_i admits the representation

$$\psi(\eta) = \frac{a_1(\eta)\xi_i^{\kappa-1} + a_2(\eta)\xi_i^{\kappa-2} + \dots + a_\kappa(\eta)}{a(\eta)} = \frac{A(\eta, \xi_i)}{a(\eta)}, \quad (2)$$

where $a(\eta), a_1(\eta), \dots, a_\kappa(\eta)$ are integral polynomials in the η , and hence $A(\eta, \xi_i)$ and $a(\eta)$ are quantities from $[\Theta]$.

The irreducible equation satisfied by y with respect to Ω_i is, by (2), of the form

$$f^{(i)}(\eta)y^{\lambda_i} + f_1^{(i)}(\eta, \xi_i)y^{\lambda_i-1} + \dots + f_{\lambda_i}^{(i)}(\eta, \xi_i) = 0,$$

and all coefficients of this equation belong to $[\Theta]$. Then the function $y_i = f^{(i)}(\eta) \cdot y$ satisfies the likewise irreducible equation, with respect to Ω_i ,

$$y_i^{\lambda_i} + f_1^{(i)}(\eta, \xi_i)y_i^{\lambda_i-1} + \dots + (f^{(i)}(\eta))^{\lambda_i-1}f_{\lambda_i}^{(i)}(\eta, \xi_i) = 0,$$

by virtue of which it depends algebraically-integrally on $[H, \xi_i]$; the same holds for the product of y_i with any polynomial in the η . In particular, the function

$$Y = f^{(0)}(\eta)f^{(1)}(\eta) \dots f^{(\sigma)}(\eta) \cdot y \quad (3)$$

depends algebraically-integrally on all domains $[H, \xi_i]$ by means of the equation, irreducible in each case with respect to Ω_i ,

$$Y^{\lambda_i} + F_1^{(i)}(\eta, \xi_i)Y^{\lambda_i-1} + \dots + F_{\lambda_i}^{(i)}(\eta, \xi_i) = 0 \quad (i = 0, 1, \dots, \sigma). \quad (4)$$

We now show that Y is the desired primitive element rationally expressible by the ϑ . Suppose, for example, that the algebraic number γ is represented by Y and $[\Theta]$:

$$\gamma = k_0(\vartheta) + k_1(\vartheta)Y + \dots + k_\nu(\vartheta)Y^\nu, \quad (5)$$

where $k_0(\vartheta), \dots, k_\nu(\vartheta)$ are polynomials from $[\Theta]$. We now determine the irreducible equation satisfied by Y with respect to the field $\mathfrak{K} = (H; k_0(\vartheta), \dots, k_\nu(\vartheta))$ arising from the coefficient domain of (5). It is known to be given by the irreducible equation of Y with respect to the intersection field of $\mathfrak{K}$ and (H, Y) , lying between (H) and (H, Y) . Now (H, Y) is identical with (H, y) by (3); and since all elements of $\mathfrak{K}$, hence also all elements of the intersection field, can be expressed rationally by finitely many ϑ , this latter field is necessarily one of the fields $\Omega_0, \Omega_1, \dots, \Omega_\sigma$, say Ω_τ . The desired irreducible equation is therefore, by (4), of degree λ_τ and is

$$Y^{\lambda_\tau} + F_1^{(\tau)}(\eta, \xi_\tau)Y^{\lambda_\tau-1} + \dots + F_{\lambda_\tau}^{(\tau)}(\eta, \xi_\tau) = 0 \quad (6)$$

with polynomials from $[\Theta]$ as coefficients. By means of (6), (5) can be rewritten in the form

$$\gamma = K_0(\vartheta) + K_1(\vartheta)Y + \dots + K_{\lambda_\tau-1}(\vartheta)Y^{\lambda_\tau-1}, \quad (7)$$

²⁸Cf. the end of § 5.

where $K_0(\vartheta), K_1(\vartheta), \dots$ belong to $\mathfrak{K}$ and, on the other hand, are polynomials from $[\Theta]$. But since γ is an algebraic number, the relation (7) does not reach degree λ_τ in Y and is therefore satisfied *identically* in Y . Hence

$$\gamma = K_0(\vartheta),$$

i.e. γ belongs to $[\Theta]$ and is consequently an algebraic integer.

As γ runs through all algebraic numbers representable by Y and $[\Theta]$, the intersection field of (H, Y) and the corresponding field $\mathfrak{K}$ always runs only through the fields $\Omega_0, \Omega_1, \dots, \Omega_\sigma$, and all the preceding arguments remain valid. That is, *by rationally integral adjunction of Y to $[\Theta]$ no algebraically fractional number can be represented*. Thus Y no longer has property 1; according to 2) or 3) it is a quantity z or ϑ , and therefore rationally representable by finitely many ϑ ; by (3), the same holds for y : *Θ has been proved to be a rational basis of all numbers*. Conversely, since every rational basis with side condition can be constructed according to 1), 2), 3) by putting Θ first in the well-ordering, we have proved: *The construction given by 1), 2), 3) leads to the most general rational basis with side condition for all numbers*.

§ 8. Division of the domains $\mathfrak{G}$ and $\mathfrak{H}$ into classes.

Alongside the question of the most general domains $\mathfrak{G}$ and $\mathfrak{H}$ there arises the further question of the *essentially different* domains, which — in a sense to be made more precise below — cannot be generated by the same construction principle. This leads to a division of these domains into classes.

Two domains will be called belonging to the same class, or isomorphic, if their elements can be put into one-to-one correspondence in such a way that the sum and product of any two elements of one domain are always assigned to the sum and product of the corresponding elements of the other. From this definition it follows that under every isomorphic relation zero and the unit correspond to themselves, and that all algebraic relations between finitely many elements are preserved for the corresponding elements, and conversely. In particular, besides the unit, all rational numbers correspond to themselves; whereas the totality of algebraic numbers — and likewise the totality of algebraic integers — goes over into itself, although an individual algebraic number can very well correspond to a conjugate.

First note that every domain from the totality $\mathfrak{M}(\mathfrak{G})$ of all domains $\mathfrak{G}$ containing a fixed algebraic basis H is isomorphic to a domain from the totality $\mathfrak{M}^*(\mathfrak{G})$ belonging to another algebraic basis H^* . For the field of all complex numbers arises from every algebraic basis by algebraic extension, and therefore has the same cardinality as the basis;²⁹ hence H and H^* have the same cardinality, and the desired isomorphic mapping is obtained by assigning the basis quantities η_i to η_i^* .³⁰ On the other hand, $\mathfrak{M}(\mathfrak{G})$ contains non-isomorphic, or essentially different, domains $\mathfrak{G}$, as follows from the examples in §§ 2, 3 and 5. For since all algebraic relations are preserved under the isomorphic mapping, an algebraic basis must again correspond to an algebraic basis; but §§ 2 and 3 showed that the functional and module domains possess no algebraic basis for which all the basis properties of the Zermelo domain would be fulfilled. Since the functional domain of § 2 is algebraically integrally closed, the subset $\mathfrak{M}(\mathfrak{H})$ also contains essentially different domains $\mathfrak{H}$.

We now show, analogously to the basis construction, how one can extract from $\mathfrak{M}(\mathfrak{G})$ a subset $\mathfrak{L}(\mathfrak{G})$ such that all domains $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$ are essentially different, while every domain from $\mathfrak{M}(\mathfrak{G})$ — and hence every domain $\mathfrak{G}$ at all — is isomorphic to a domain from $\mathfrak{L}(\mathfrak{G})$. Subject $\mathfrak{M}(\mathfrak{G})$ to a well-ordering Ω ; then a domain $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ may or may not be isomorphic to some preceding one. The domains $\mathfrak{G}$ isomorphic to no preceding one form a set $\mathfrak{L}(\mathfrak{G})$, which by construction contains only essentially different domains $\mathfrak{G}$. The induction argument also shows that every domain from $\mathfrak{M}(\mathfrak{G})$ is isomorphic to a domain from $\mathfrak{L}(\mathfrak{G})$. For if $\mathfrak{G}_1$ were the first domain for which this failed, then $\mathfrak{G}_1$ either belongs to $\mathfrak{L}(\mathfrak{G})$, or is isomorphic to a preceding domain $\mathfrak{G}_0$, which by assumption is isomorphic to a domain from $\mathfrak{L}(\mathfrak{G})$; hence the same would hold for $\mathfrak{G}_1$. Since every domain $\mathfrak{G}$ belongs to some set $\mathfrak{M}^*(\mathfrak{G})$ and is therefore isomorphic to a domain from $\mathfrak{M}(\mathfrak{G})$, *the totality of the classes of domains $\mathfrak{G}$ is represented by $\mathfrak{L}(\mathfrak{G})$* .

If, in a domain $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$, one replaces the basis quantities η by those η^* of some other basis, then, as shown above, a domain isomorphic to $\mathfrak{G}$ arises. Conversely, let $\mathfrak{G}^*$ be any domain and suppose it is isomorphic to a domain $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$. If the basis quantities η correspond to the quantities η^* from $\mathfrak{G}^*$, then these again form an algebraic basis, and $\mathfrak{G}^*$ contains the same algebraic functions of η^* as $\mathfrak{G}$ contains of η —

²⁹Steinitz, §§ 23 and 24.

³⁰This conclusion of course does not hold for the sets $\mathfrak{N}(\mathfrak{G})$ and $\mathfrak{N}^*(\mathfrak{G})$ belonging to two rational bases Θ and Θ^* ; nor can the isomorphism of $\mathfrak{N}(\mathfrak{G})$ and $\mathfrak{N}^*(\mathfrak{G})$ be inferred from the mutual rational expressibility of Θ and Θ^* . What is obtained, however, is an isomorphic relation between the fields (Θ) and (Θ^*) .

or their conjugates. Thus every domain isomorphic to $\mathfrak{G}$ arises by letting the algebraic basis H run through every possible algebraic basis in $\mathfrak{G}$ and in its conjugates with respect to $(H)^{31}$ — if these differ from $\mathfrak{G}$. From each fixed domain $\mathfrak{G}_i$ in $\mathfrak{L}(\mathfrak{G})$ one obtains in this way the class $\mathfrak{R}_i$, which contains all and only the domains $\mathfrak{G}$ isomorphic to $\mathfrak{G}_i$, although possibly each domain occurs several times or even infinitely often. From $\mathfrak{R}_i$ one can again, by the procedure above, extract a subset $\mathfrak{R}_i^*$ which contains every domain isomorphic to $\mathfrak{G}_i$ once and only once. As $\mathfrak{R}_i^*$ runs through all classes, one obtains the totality of all domains $\mathfrak{G}$, every domain once and only once. To characterize the class $\mathfrak{R}_i$, one can of course use, instead of $\mathfrak{G}_i$, any other domain of the class from which all domains of $\mathfrak{R}_i$ can be derived as above. In summary:

From $\mathfrak{M}(\mathfrak{G})$ one can extract a subset $\mathfrak{L}(\mathfrak{G})$ such that the domains $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$ are in one-to-one correspondence with the totality of classes (without repetition). From $\mathfrak{G}_i$ the corresponding class $\mathfrak{R}_i$ arises by letting H run through every possible algebraic basis in $\mathfrak{G}_i$ and its conjugates with respect to (H) . If $\mathfrak{R}_i^$ denotes all mutually distinct domains from $\mathfrak{R}_i$, and $\mathfrak{R}_i^*$ runs through all classes, then the totality of all domains $\mathfrak{G}$ arises, every domain once and only once.*³²

The analogous statement holds for the domains $\mathfrak{H}$ made from algebraically integral transcendental numbers.

§ 9. An example of countably infinitely many classes of domains $\mathfrak{G}$.

The cardinality of the constructions given in § 4 b) and § 6 b) can be read off from the remarks thereto as equal to the cardinality of all well-orderings, hence as 2^c , if c denotes the cardinality of the continuum. But from this no conclusion can be drawn about the cardinality of the totality of all domains $\mathfrak{G}$, counted once and only once, and still less about the cardinality of all classes. That the number of classes is certainly not finite will now be shown by an example of countably infinitely many domains $\mathfrak{G}$ — analogous to those of § 5 — which will all turn out to be essentially different and hence supply countably infinitely many classes.³³

Analogously to § 5, (1), let the domain $\mathfrak{G}_\sigma$ consist of the totality of quantities

$$z = a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta) \pmod{\mathfrak{M}_\sigma(\eta)}, \quad (1)$$

where $a_i(\eta)$ denotes homogeneous forms of i -th dimension in the η ; and the module $\mathfrak{M}_\sigma$ contains as basis all power products of σ -th dimension in the η ³⁴ and as multipliers all algebraic-integer functions of the η . We show that under an isomorphic relation between two domains $\mathfrak{G}_\sigma$ and $\mathfrak{G}_\tau$ ($\sigma \neq \tau$) the modules $\mathfrak{M}_\sigma$ and $\mathfrak{M}_\tau$ must necessarily correspond isomorphically; the impossibility then follows easily.

Let the indeterminates in $\mathfrak{G}_\tau$ be denoted by ξ ; under the isomorphic relation, let the quantities $f(\eta)$ from $\mathfrak{G}_\sigma$ correspond to these ξ . Since the isomorphic relation is preserved under forming quotients, the domain $\mathfrak{G}_\xi$ of all algebraic-integer functions of the ξ corresponds to an algebraically integrally closed domain $\mathfrak{T}$ consisting of all functions algebraically integral over the $f(\eta)$, and hence also algebraically integral in the η .

Suppose (1) corresponds to a quantity from $\mathfrak{M}_\tau$. Then, by the module property of $\mathfrak{M}_\tau$, the product of (1) with an arbitrary quantity $\psi(\eta)$ from $\mathfrak{T}$ must also belong to $\mathfrak{G}_\sigma$; in formulas:

$$(a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta))\psi(\eta) \equiv b_0 + b_1(\eta) + \cdots + b_{\sigma-1}(\eta) \pmod{\mathfrak{M}_\sigma}. \quad (2)$$

Setting all η equal to zero gives $a_0\psi(0) = b_0$, so (2) becomes

$$(a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta))(\psi(\eta) - \psi(0)) \equiv c_1(\eta) + \cdots + c_{\sigma-1}(\eta) \pmod{\mathfrak{M}_\sigma}. \quad (3)$$

Now, besides $\psi(\eta)$, the functions

$$\chi_\nu(\eta) = \sqrt[\nu]{\psi(\eta) - \psi(0)}$$

also belong to $\mathfrak{T}$ for every ν ; and since $\chi_\nu(0) = 0$, the formulas analogous to (3) are:

$$(a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta))\chi_\nu(\eta) \equiv c_1^{(\nu)}(\eta) + \cdots + c_{\sigma-1}^{(\nu)}(\eta) \pmod{\mathfrak{M}_\sigma} \quad (\nu = 1, 2, \dots). \quad (4)$$

³¹The elements of these domains conjugate to $\mathfrak{G}$ can certainly be expressed rationally by the elements of $\mathfrak{G}$ by II, but not necessarily integrally and rationally; the conjugates can therefore differ from $\mathfrak{G}$.

³²Here $\mathfrak{M}(\mathfrak{G})$ is to be constructed according to §§ 6 and 7: by § 7 one extends an algebraic basis H to every rational basis Θ with side condition arising from it, and for each Θ forms, by § 6 b), the totality $\mathfrak{M}(\mathfrak{G})$ of all domains $\mathfrak{G}$ containing Θ .

³³The domains $\mathfrak{G}$ of § 6 likewise give countably infinitely many classes, but the proof is somewhat more complicated.

³⁴Naturally, in each individual z only finitely many power products of the η occur in the module.

Let $A(\eta)$, of degree λ , denote the product of $\chi_1 = \psi(\eta) - \psi(0)$ with its $\kappa - 1$ conjugates. Since $\chi_1(0) = 0$, $A(\eta)$ must begin with terms of at least first dimension. From (4) one obtains

$$a_0\chi_\nu(\eta) \equiv 0 \pmod{\mathfrak{M}_1}; \quad a_0^\nu\chi_1(\eta) \equiv 0 \pmod{\mathfrak{M}_\nu}; \quad a_0^{\nu\kappa}A(\eta) \equiv 0 \pmod{\mathfrak{M}_{\nu\kappa}}, \quad (5)$$

and therefore, for $a_0 \neq 0$, as in § 3, $\lambda > \nu\kappa$. But since $\nu < \frac{\lambda}{\kappa}$ contradicts the algebraically integral closedness of $\mathfrak{T}$, necessarily $a_0 = 0$. From (4) one now obtains

$$a_1(\eta)\chi_\nu(\eta) \equiv c_1^{(\nu)}(\eta) \pmod{\mathfrak{M}_2}; \quad [a_1(\eta)]^{\nu\kappa}A(\eta) \equiv [c_1^{(\nu)}(\eta)]^{\nu\kappa} \pmod{\mathfrak{M}_{\nu\kappa+1}}, \quad (6)$$

and from this, since $A(\eta)$ begins with terms of at least first dimension, $c_1^{(\nu)}(\eta) = 0$. Thus, in complete analogy with (5), for every ν :

$$a_1(\eta)\chi_\nu(\eta) \equiv 0 \pmod{\mathfrak{M}_2}; \quad [a_1(\eta)]^{\nu\kappa}A(\eta) \equiv 0 \pmod{\mathfrak{M}_{2\nu\kappa}},$$

and hence $a_1(\eta) = 0$. Continuing in this way, the alternating application of (5) and (6) gives

$$a_0 = 0, \quad a_1(\eta) = 0, \quad \dots, \quad a_{\sigma-1}(\eta) = 0.$$

Since the analogous consideration also holds with respect to $\mathfrak{G}_\tau$: *Under every isomorphic relation between $\mathfrak{G}_\sigma$ and $\mathfrak{G}_\tau$ ($\sigma \neq \tau$), the modules $\mathfrak{M}_\sigma$ and $\mathfrak{M}_\tau$ correspond isomorphically.*

Suppose, for instance, that $\sigma > \tau$. To the indeterminates ξ there corresponded the quantities $f(\eta)$ from $\mathfrak{G}_\sigma$; and, since all quantities from $\mathfrak{G}_\tau$ can be represented by polynomials in the ξ mod. $\mathfrak{M}_\tau$, the correspondence of $\mathfrak{M}_\tau$ and $\mathfrak{M}_\sigma$ implies that all quantities from $\mathfrak{G}_\sigma$ are obtained as polynomials in the $f(\eta)$ mod. $\mathfrak{M}_\sigma$. Since $\sigma > \tau \geq 1$, the indeterminates η certainly do not belong to $\mathfrak{M}_\sigma$, and hence, being integrally and rationally expressible by the $f(\eta)$ mod. $\mathfrak{M}_\sigma$, there must be an $f(\eta)$ with non-vanishing linear terms. Thus let

$$\begin{aligned} \xi : f(\eta) &\equiv a_0 + a_1(\eta) \pmod{\mathfrak{M}_2} \quad [a_1(\eta) \neq 0], \\ \xi^\tau : [f(\eta)]^\tau &\equiv a_0^\tau \pmod{\mathfrak{M}_1} \quad \text{for } a_0 \neq 0, \\ &\equiv [a_1(\eta)]^\tau \pmod{\mathfrak{M}_{\tau+1}} \quad \text{for } a_0 = 0. \end{aligned}$$

Now ξ^τ belongs to $\mathfrak{M}_\tau$, while $[f(\eta)]^\tau$ begins with non-vanishing terms of zeroth or τ -th dimension and, since $\tau < \sigma$, cannot belong to $\mathfrak{M}_\sigma$, contrary to the correspondence of $\mathfrak{M}_\sigma$ and $\mathfrak{M}_\tau$ derived above. The case $\tau > \sigma$ is dealt with at the same time by interchanging η and ξ . It is therefore shown that the domains $\mathfrak{G}_\sigma$ and $\mathfrak{G}_\tau$ are *not isomorphic, or essentially different*. Thus the domains $\mathfrak{G}_i$ ($i = 1, 2, \dots$ in inf.) give an example of countably infinitely many classes of domains $\mathfrak{G}$.

Remark. Although any two domains $\mathfrak{G}_i$ are non-isomorphic, it can very well happen, for two domains, that each is isomorphic to a part of the other.³⁵ Let, for example, $\tau = 1$ and σ be arbitrary:

$$\mathfrak{G}_1 : a_0 + \mathfrak{M}_1(\xi); \quad \mathfrak{G}_\sigma : a_0 + a_1(\eta) + \dots + a_{\sigma-1}(\eta) + \mathfrak{M}_\sigma(\eta).$$

The assignment $\xi = \eta$ makes $\mathfrak{G}_\sigma$ a proper divisor of $\mathfrak{G}_1$; conversely, the mutually one-to-one assignment in $\mathfrak{G}_1$ and $\mathfrak{G}_\sigma$ given by $\xi = \eta^\sigma$, $\eta = \sqrt[\sigma]{\xi}$ makes $\mathfrak{G}_1$ a proper divisor of $\mathfrak{G}_\sigma$. Thus every $\mathfrak{G}_i$ is also isomorphic to a proper divisor of itself. In contrast with the notion of intersection, the notion of an intersection class — that is, a class such that each of its domains would be isomorphic to a proper divisor of every domain of every other class — does not exist, at least not as a uniquely determined notion.

§ 10. The integral quantities of an arbitrary field.

The developments of the preceding paragraphs have, for the domains $\mathfrak{G}$ — provided that the algebraic numbers are not included in the defining properties III and IV for the domains $\mathfrak{G}$ — relied only on the fact that the totality of all complex numbers forms a field;³⁶ for the domains $\mathfrak{H}$, they relied on the further fact that this field is algebraically closed. The results obtained therefore directly admit the *generalization to the most general field*, which, following Weber and E. Steinitz,³⁷ is abstractly defined as “a system of elements

³⁵This behavior can also occur for fields; cf. Steinitz, loc. cit., conclusion.

³⁶Only in § 7 was the further assumption used that the “prime field” of the rational numbers contains infinitely many elements; in the case of a prime field with only finitely many elements, however, this paragraph simply disappears, as we shall see.

³⁷loc. cit., introduction.

with two operations, addition and multiplication, which are subject to the associative and commutative laws, are connected by the distributive law, and admit unrestricted and unique inverse operations.”³⁸

Every such field contains a unit element, and consequently the “prime field” derived from it by the operations of addition and multiplication and their inverses. This prime field is either of the type of the rational numbers (characteristic 0), or of the type of the residue-class system of a prime number p (characteristic p).³⁹ The *integral quantities of the prime field* are then well-defined as the quantities arising from the unit element by addition and subtraction; for prime fields of characteristic p , the integral quantities already exhaust the prime field, and there are no fractional quantities of the prime field. Every algebraically closed field contains an “absolutely algebraic field,” arising from the prime field by adjoining all elements algebraic over the prime field; for fields of characteristic 0 this is therefore of the type of the field of all algebraic numbers. The *algebraically integral quantities of the absolutely algebraic field* are then well-defined as all quantities algebraically integral over the integral quantities of the prime field (or, what is the same thing, over the unit); for fields of characteristic p there are therefore also no algebraically fractional quantities of the absolutely algebraic field. After these preliminary remarks, the definition of the “integral” and “algebraically integral” quantities can be transferred to arbitrary fields, respectively to algebraically closed fields: *The integral quantities of a field $\mathfrak{R}$ are defined as an integral domain $\mathfrak{G}$ of quantities from $\mathfrak{R}$ whose quotient field⁴⁰ exhausts $\mathfrak{R}$, and which contains the unit element but no fractional quantity of the prime field.*

The algebraically integral quantities of an algebraically closed field $\mathfrak{R}$ are defined as an algebraically integrally closed integral domain $\mathfrak{H}$ of quantities from $\mathfrak{R}$ whose quotient field exhausts $\mathfrak{R}$, and which contains the unit element but no algebraically fractional quantity of the absolutely algebraic field.

For fields of characteristic zero, the results obtained in the preceding paragraphs apply literally, once one merely replaces the “algebraic numbers” by the quantities of the prime field, respectively of the absolutely algebraic field. For fields of characteristic p , the results simplify still further because the prime field contains no fractional quantities, so that the side condition for the rational basis and for the systems to be adjoined simply drops away.⁴¹ Thus the result is: *The abstract definition permits as integral quantities of a field of characteristic p : every integral domain derived from an arbitrary rational basis, the field itself, and every integral domain lying between these domains and the field. Correspondingly, as algebraically integral quantities of an algebraically closed field of prime characteristic one has: every algebraically integrally closed integral domain lying between the domain derived from an algebraic basis and the field itself — including the two boundary domains.* For the most general fields of prime characteristic, therefore, as with the corresponding prime fields, the distinction between integral and fractional is effaced.

Erlangen, 30 March 1915.

³⁸Only division by zero is to be excluded.

³⁹Steinitz, § 4.

⁴⁰That is, the field consisting of all quotients of pairs of quantities from $\mathfrak{G}$.

⁴¹Cf. the first note to this paragraph.